



Requirement & Design Specification
Crowndine - Restaurant Management System

Subject: SWD392

Version: 1.0

Record of Changes

Version	Date	In Charge	Change Description
1.0	July 2026	Project Team	Initial version

* A - Added M - Modified D - Deleted

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I Requirement Specification

I.1 Problem Description

In today's Food and Beverage (F&B) industry, restaurants continue to face challenges in optimizing operational processes, including table reservations, order management, kitchen coordination, and payment handling, due to the lack of an integrated and efficient management system.

The main problems are described as follows:

- **Inefficient reservation and ordering experience:** Customers often have limited flexibility when selecting their preferred tables and placing orders in advance. As a result, restaurants face difficulties in preparing ingredients, allocating seating arrangements, and anticipating customer demand effectively.
- **Operational communication challenges:** Information exchange between service staff, kitchen staff, and cashiers can be slow or prone to errors. Miscommunication may lead to incorrect orders, delays in food preparation, and reduced service quality, particularly during peak business hours.
- **Limited decision-support capabilities:** Restaurant managers often struggle to consolidate and analyze business data from multiple sources. This makes it difficult to identify customer trends, optimize menu offerings, improve inventory management, and make data-driven business decisions.
- **Inefficient resource and inventory management:** Without a centralized system, monitoring ingredient consumption, tracking stock levels, and managing restaurant resources can be time-consuming and inaccurate, potentially leading to waste and increased operational costs.

CrownDine is designed to address these challenges through a centralized restaurant management platform that streamlines reservation, ordering, kitchen coordination, payment processing, and business management activities. Additionally, the system integrates Artificial Intelligence (AI) to provide valuable insights and support smarter business decisions, helping restaurants improve operational efficiency and enhance customer satisfaction.

I.2 Major Features

The CrownDine system provides separate feature groups for customers, staff, and administrators. These features are adapted from the implemented business logic in both the Backend and Frontend, including reservation workflows, online payment, restaurant operations, management dashboards, and AI chatbot integrations.

I.2.1 Features for Customers

F01: Viewing Menu & Product Details

- Browse food items and drinks grouped by categories.

- Browse food combo packages.
- View detailed information of items and combos, including descriptions, prices, images, and customer feedback.
- Search items and combos by name and filter them by categories.
- Add items to the favorites list, view favorite items, and manage the favorites list.

F02: Smart Table Reservation

- Use a step-by-step reservation wizard.
- Select booking date, dining time slot, and guest count.
- View an interactive table map showing table locations, floor layouts, table status, shape, and seating capacity.
- Select preferred tables for the reservation.

F03: Pre-order Cart Management

- Add food items and combos to the reservation order during booking.
- View the food reservation cart.
- Edit cart items by changing quantities or removing items.

F04: Booking Placement & Payment

- Review reservation details and total price before checkout.
- Apply discount vouchers.
- Make online payments through PayOS integration using QR Code or bank transfer.
- Receive real-time payment success or failure redirection and booking confirmation.
- Cancel pending or upcoming bookings.

F05: Customer Authentication

- Register a new account with email verification via OTP.
- Log in using username and password or Google OAuth2.
- Securely log out, refresh login tokens, and update passwords.

F06: Profile & Loyalty Program

- View and update customer personal profile details.
- Request email update with OTP verification for enhanced security.
- Upload profile avatar images processed and hosted through Cloudinary.
- View reservation history, order details, and statuses.
- Track accumulated loyalty membership points.
- Exchange loyalty points for discount vouchers.

F07: Customer AI Booking Assistant

- Chat in real time with AI chatbot for booking assistance.
- Ask questions about the menu, dishes, prices, and opening hours.
- Automatically parse relative time expressions into actual calendar dates.
- View available table recommendations matching guest counts.
- Confirm table selection and receive direct routing links to step 3 for pre-order menu or step 4 for payment and checkout.

I.2.2 Features for Staff

F08: Table Reservation Processing

- View the full list of customer reservations and filter by date ranges and reservation statuses, including Pending, Confirmed, Checked-in, Completed, and Cancelled.
- Look up booking details and search reservations.
- Perform customer check-in when guests arrive at the restaurant.
- Reassign or update tables for an active reservation.
- Cancel reservation bookings for customers with required cancellation reasons.

F09: Cashier & Walk-in Dine-in Orders

- Create walk-in dine-in orders directly for a table.
- Retrieve customer profiles by phone number to link them to orders for point accumulation.
- Append items, modify quantities, or remove items in active dine-in orders.
- Apply vouchers to active orders.
- Merge tables or split bills.
- Generate PayOS checkout QR codes for bills, process payments, and track transactions.
- Print invoice receipts and close completed tables by setting table status to Free.

F10: Kitchen Display System & Order Processing

- Use a real-time Kitchen Display System (KDS) for kitchen staff to view pending and preparation order queues.
- Group identical ordered dishes into batch preparations.
- Update item status in real time from Pending to Preparing to Ready using WebSockets.
- Automatically notify floor staff through toast notifications when dishes are marked as Ready for serving.

F11: Shift Schedule & Attendance Tracking

- View assigned weekly work schedules.
- View personal attendance history.
- Perform check-in and check-out to log work hours.

I.2.3 Features for Admin

F12: Restaurant Layout & Table Design

- Add, edit, and delete floors and areas.
- Add, edit, and delete restaurant tables.
- Configure the visual layout by editing table names, coordinates, width, height, shape, and seat capacity to build an interactive floor plan.

F13: Menu & Category Management

- Add, edit, delete, and search food categories.
- Add, edit, delete, and search food items and combo packages.

- Manage item pricing, upload dish images through Cloudinary integration, write descriptions, and update combo item lists.

F14: Promotion & Voucher Management

- Create, edit, and delete discount vouchers.
- Configure voucher types, including percentage discounts and fixed amounts, minimum order values, expiry dates, usage limits, and descriptions.
- Manually assign vouchers to specific users.
- Manage point exchange rates and point costs for vouchers.

F15: Staff Account & Role Assignment

- Create new staff accounts.
- Search staff profiles by name and view detailed staff information.
- Update staff profiles, edit status as Active or Blocked, and assign system roles and authorities such as Admin, Staff, Kitchen, and Cashier.

F16: Employee Shift Scheduling

- Configure daily shifts, such as Morning, Afternoon, and Evening.
- Assign weekly work schedules to staff members.
- Update or reassign scheduled shifts, change work schedule status, and delete schedules.

F17: Attendance Auditing & Payroll Summary

- View weekly schedules and clock-in statuses for all staff.
- Track detailed clock-in logs.
- View monthly attendance summaries.

F18: Sales Analytics Dashboard & Excel Export

- View interactive dashboard sales graphs, including revenue by hour or day, customer counts, and growth rates.
- Export comprehensive sales reports to Excel spreadsheets.

I.3 System Context

The system context of the CrownDine Restaurant Management System illustrates the interactions between the system and its entities, including external users and external systems.

External Users:

- **Customer:** The system's end-user who can browse the digital menu, view food details, make table reservations, pre-order dishes, make payments, track booking status, manage personal profiles, and use the AI booking assistant.
- **Staff:** Responsible for handling restaurant operations, including managing reservations, checking in customers, creating walk-in orders, updating order information, processing bills, merging tables, splitting invoices, managing Kitchen Display System (KDS) orders, updating dish preparation status, and notifying when dishes are ready to serve.

- **Admin:** Manages the entire system, including restaurant layout, tables, menu items, categories, staff accounts, user roles, vouchers, staff schedules, attendance records, revenue reports, and AI-based business analytics.

External Systems:

- **PayOS Payment Gateway:** Handles online payment transactions through QR Code and bank transfer, and returns payment status to the CrownDine system.
- **Google Authentication Service:** Provides login services using users' Google accounts through Google OAuth2 authentication.
- **Google Gemini AI Service:** Provides AI chatbot and data analysis functions, including revenue analysis, popular dish analysis, customer trend analysis, and demand prediction support.
- **Cloudinary Image Storage Service:** Stores and delivers uploaded images, including food images, combo images, and user avatar images.
- **Mailing / Notification System:** Sends OTP verification, booking confirmations, payment notifications, promotional messages, and real-time operational alerts to users.

I.4 Non-Functional Requirements

Non-functional requirements describe the quality attributes and operating constraints that the Crowndine Restaurant Management System must satisfy. These requirements ensure that the system is not only functionally correct but also practical, secure, reliable, and maintainable in a real restaurant environment.

- **Usability:** The system shall provide an intuitive, seamless experience for all roles. For customers, the 4-step booking journey, interactive Table Map, and vibrant digital menu must be highly accessible. For staff and kitchen personnel, the Calendar View and KDS interfaces must be visually clear to manage real-time operations without requiring extensive technical training.
- **Reliability:** CrownDine must guarantee absolute accuracy in preserving reservations, order batching sequences, and financial records processed via PayOS (QR Code and Bank Transfer). The system must consistently maintain state synchronization across all modules and reliably execute the automated cancellation and 80% refund policy without any data loss or manual intervention.
- **Performance:** The platform shall process core actions—such as interactive table selection, cart updates, and payment authorizations—with minimal latency. The Kitchen Display System (KDS) must efficiently utilize WebSockets to ensure instant, real-time status updates under heavy loads, while the integrated Gemini AI chatbot must deliver prompt analytical insights.
- **Security:** All user data and transactions shall be strictly protected. The system must implement robust authentication utilizing JWT and Google OAuth2. Furthermore, a granular role-based access control (RBAC) mechanism must precisely isolate permissions among Admin, Staff, Kitchen, and Customer roles to prevent unauthorized system modifications.

- **Maintainability:** The software architecture shall be highly modular, enabling developers to seamlessly implement updates to the WebSockets infrastructure, PayOS integrations, and AI features without system downtime. This ensures easy future expansion and allows administrators to flexibly configure business rules via the admin panel rather than deep code changes.

I.5 Functional Requirements

FR1. The system shall allow customers to view menu items and restaurant information.

FR2. The system shall allow customers to create and manage reservations.

FR3. The system shall allow customers to place and track orders.

FR4. The system shall allow staff to confirm, update, and complete orders.

FR5. The system shall allow administrators to manage users, menu items, and reports.

I.5.1 Use Case Diagrams

Based on the described user requirements and detailed analysis, the CrownDine Restaurant Management System is deployed on a web platform with role-based access for authentication, customers, staff, and administrators. The following diagrams include the four overview modules and detailed use case modules for key customer, staff, and administration workflows.

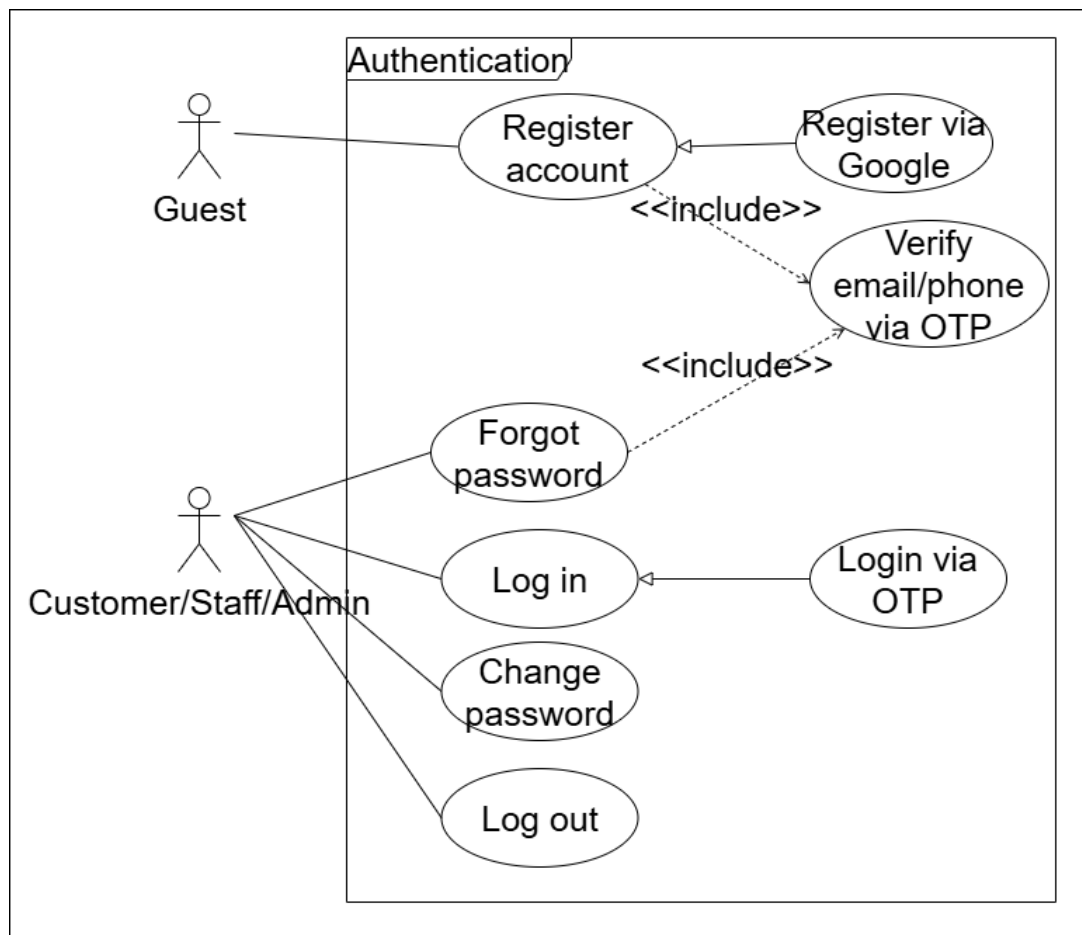


Figure I-1: Overview Use Case Diagram for Authentication

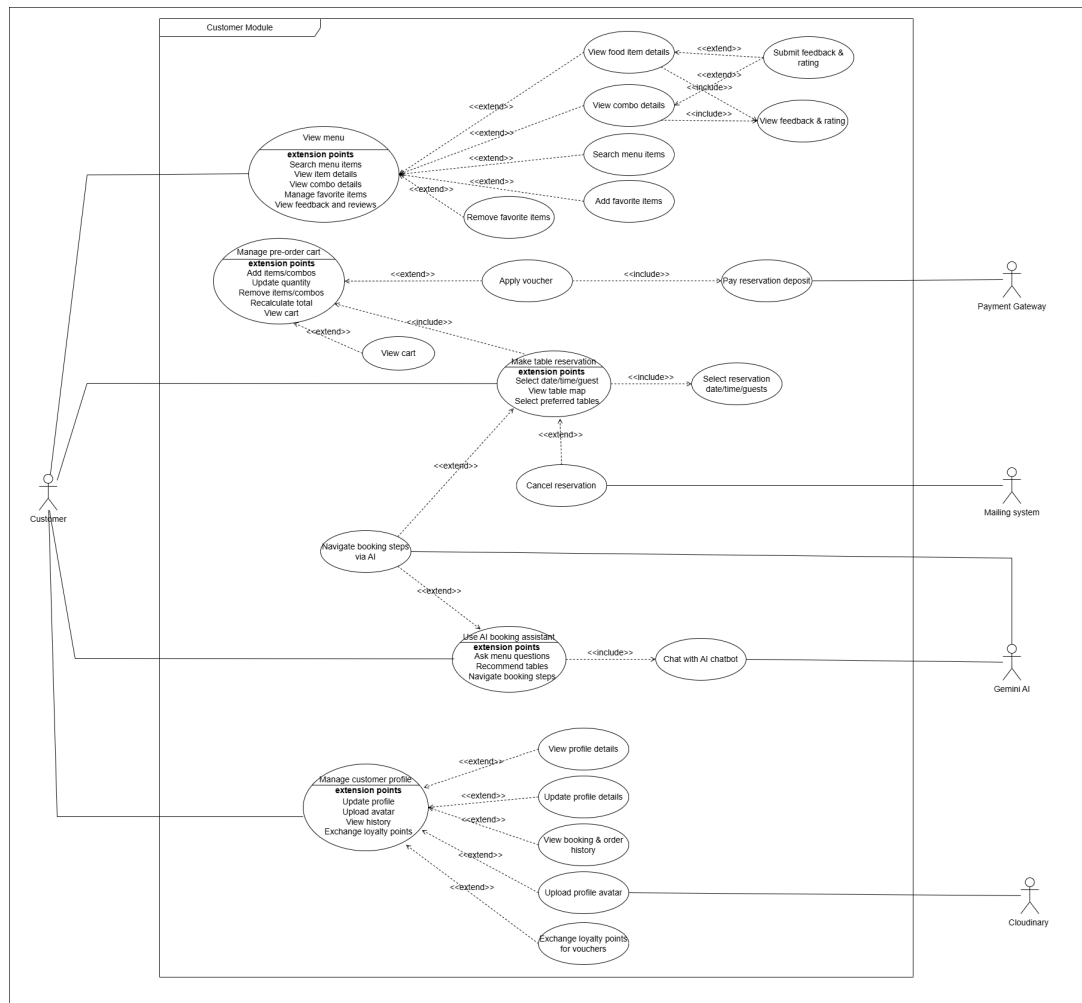


Figure I-2: Overview Use Case Diagram for Customer Site

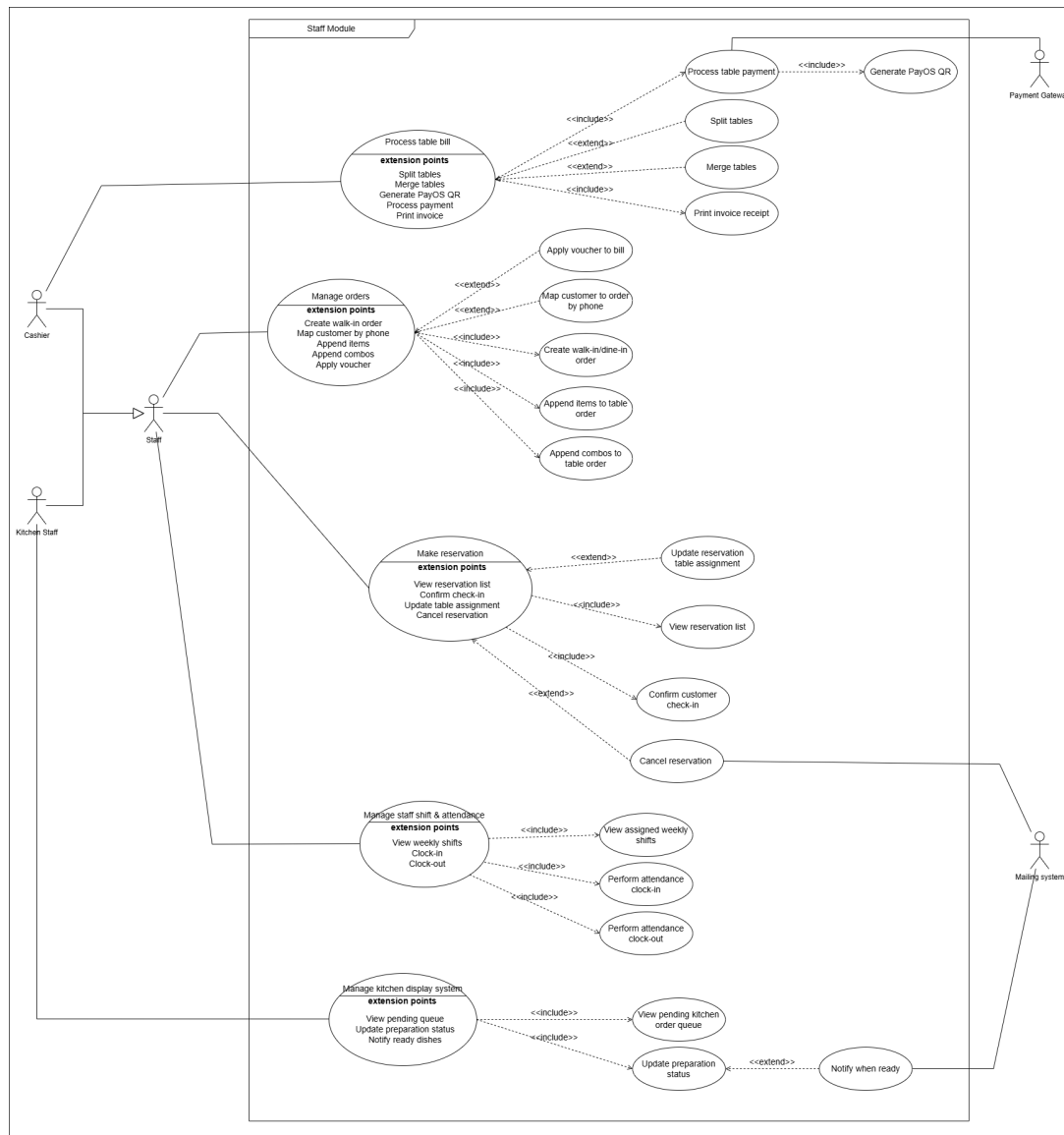


Figure I-3: Overview Use Case Diagram for Staff Site

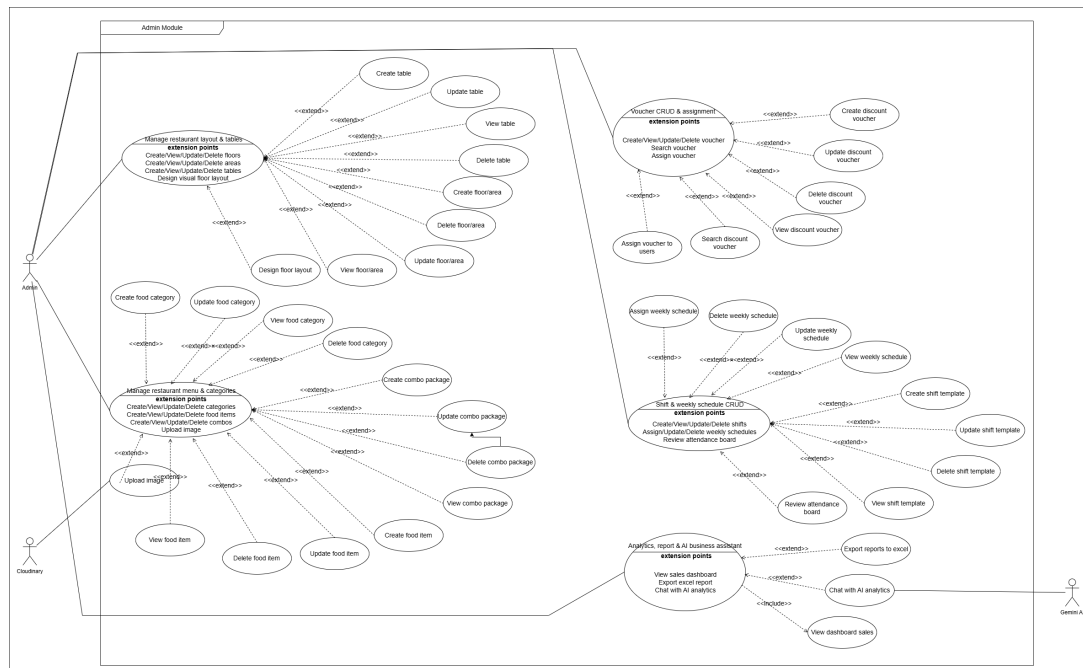


Figure I-4: Overview Use Case Diagram for Administration Site

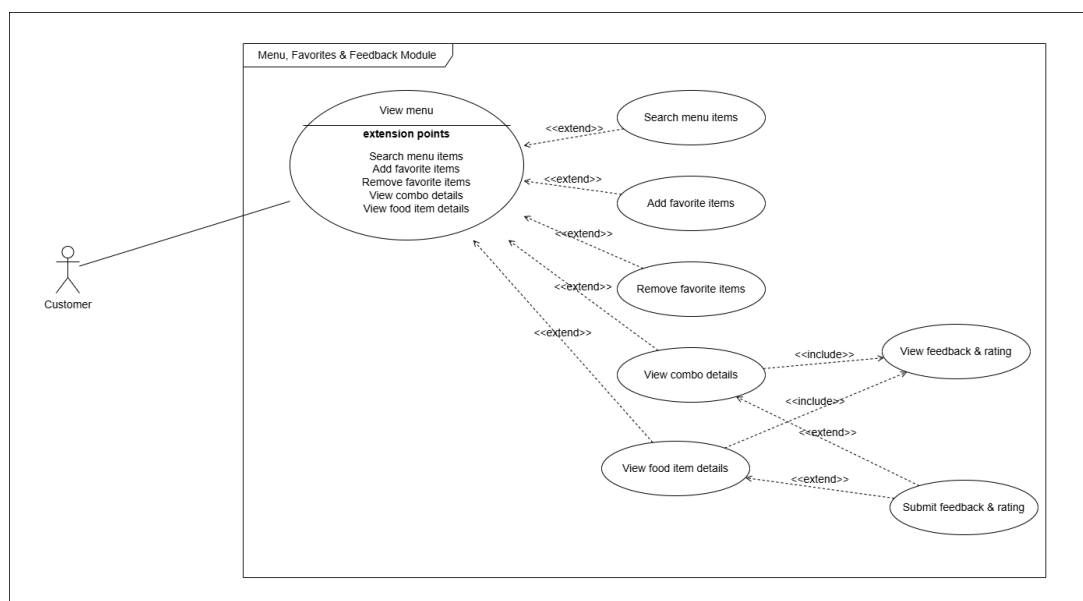


Figure I-5: Detailed Use Case Diagram for Viewing Menu, Favorites and Feedback

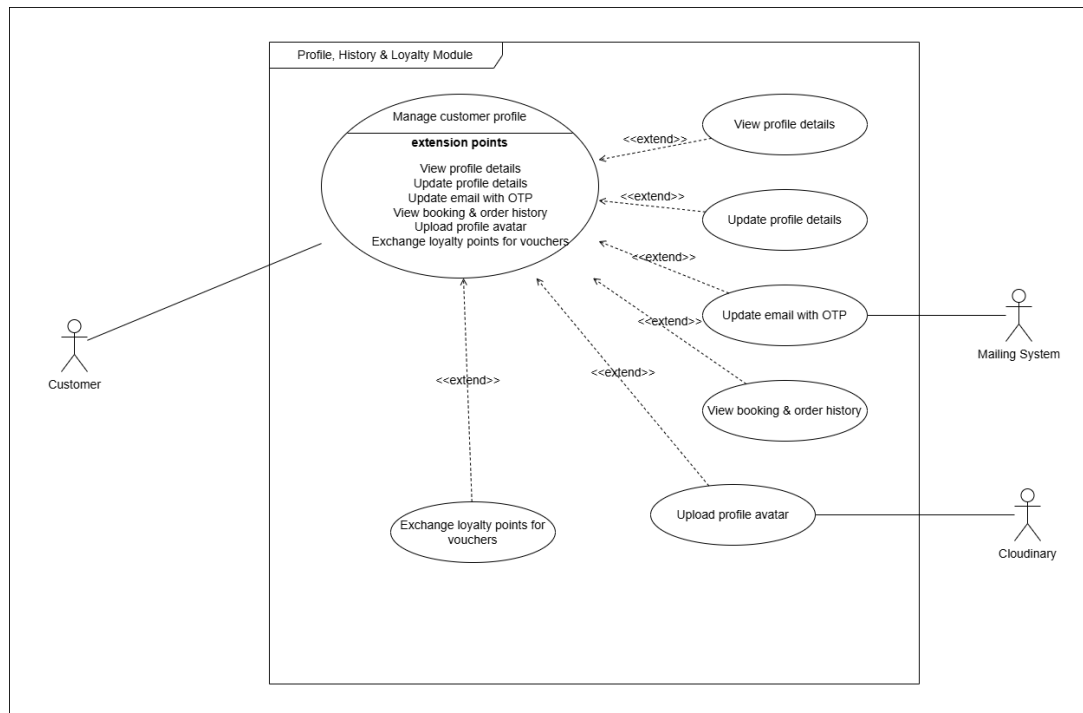


Figure I-6: Detailed Use Case Diagram for Managing Profile, History and Loyalty

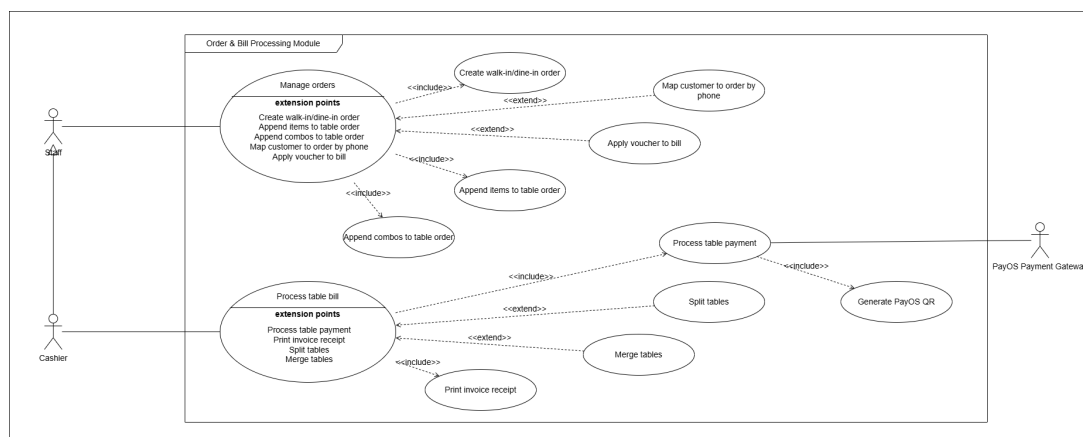


Figure I-7: Detailed Use Case Diagram for Order and Bill Processing

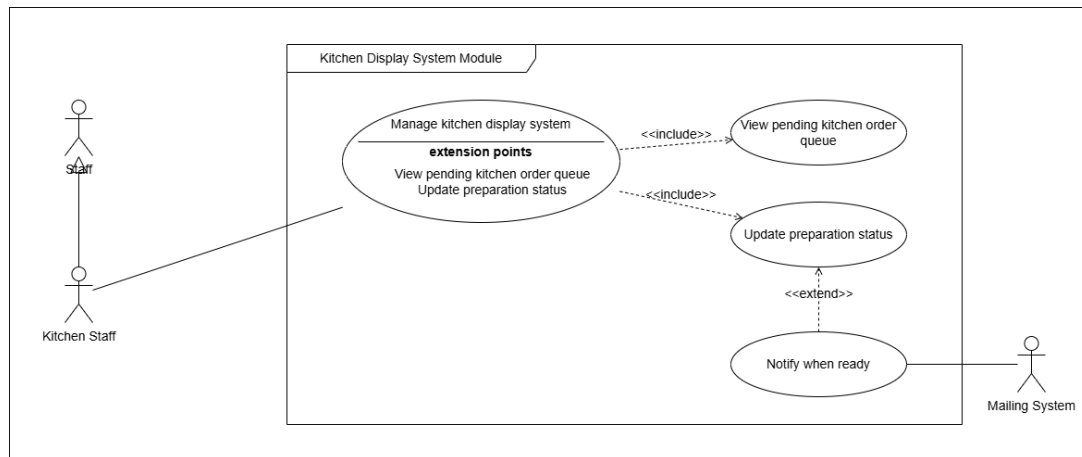


Figure I-8: Detailed Use Case Diagram for KDS and Order Preparation

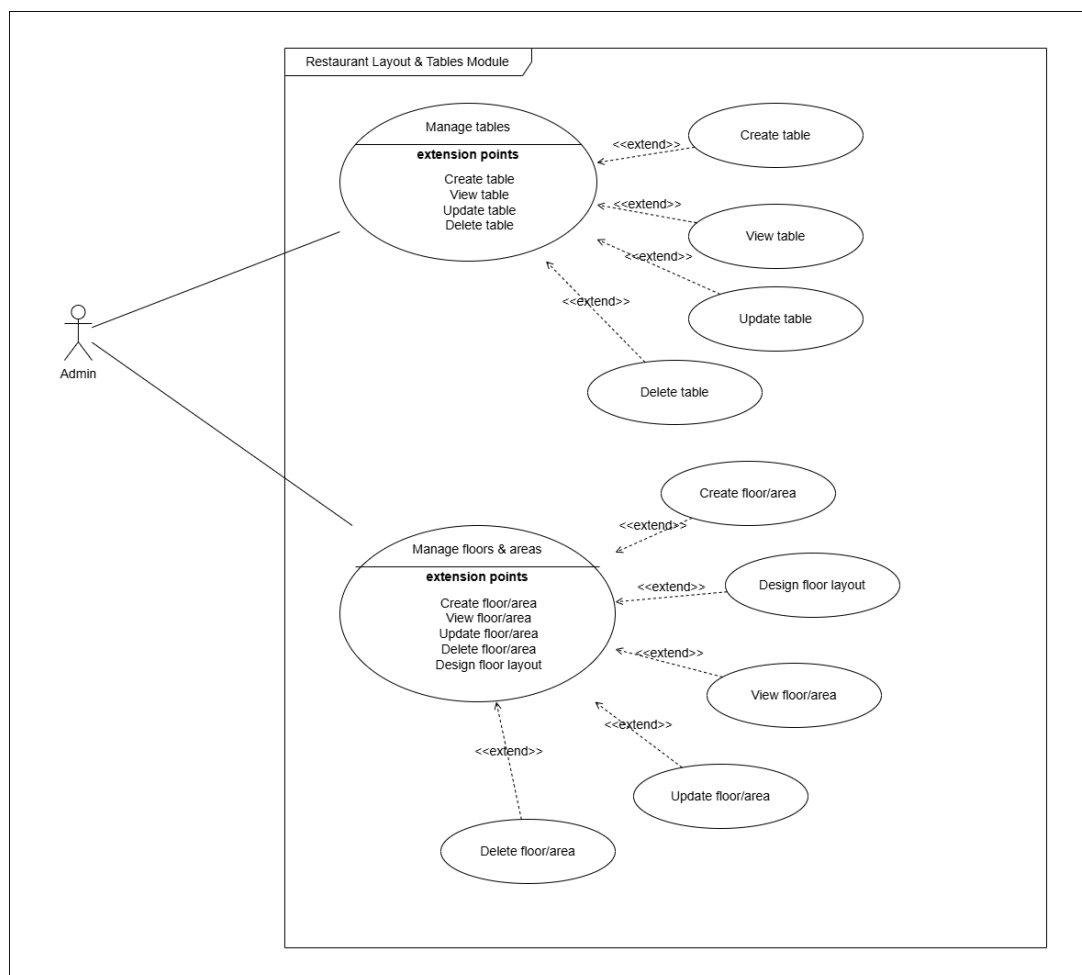


Figure I-9: Detailed Use Case Diagram for Managing Restaurant Layout and Tables

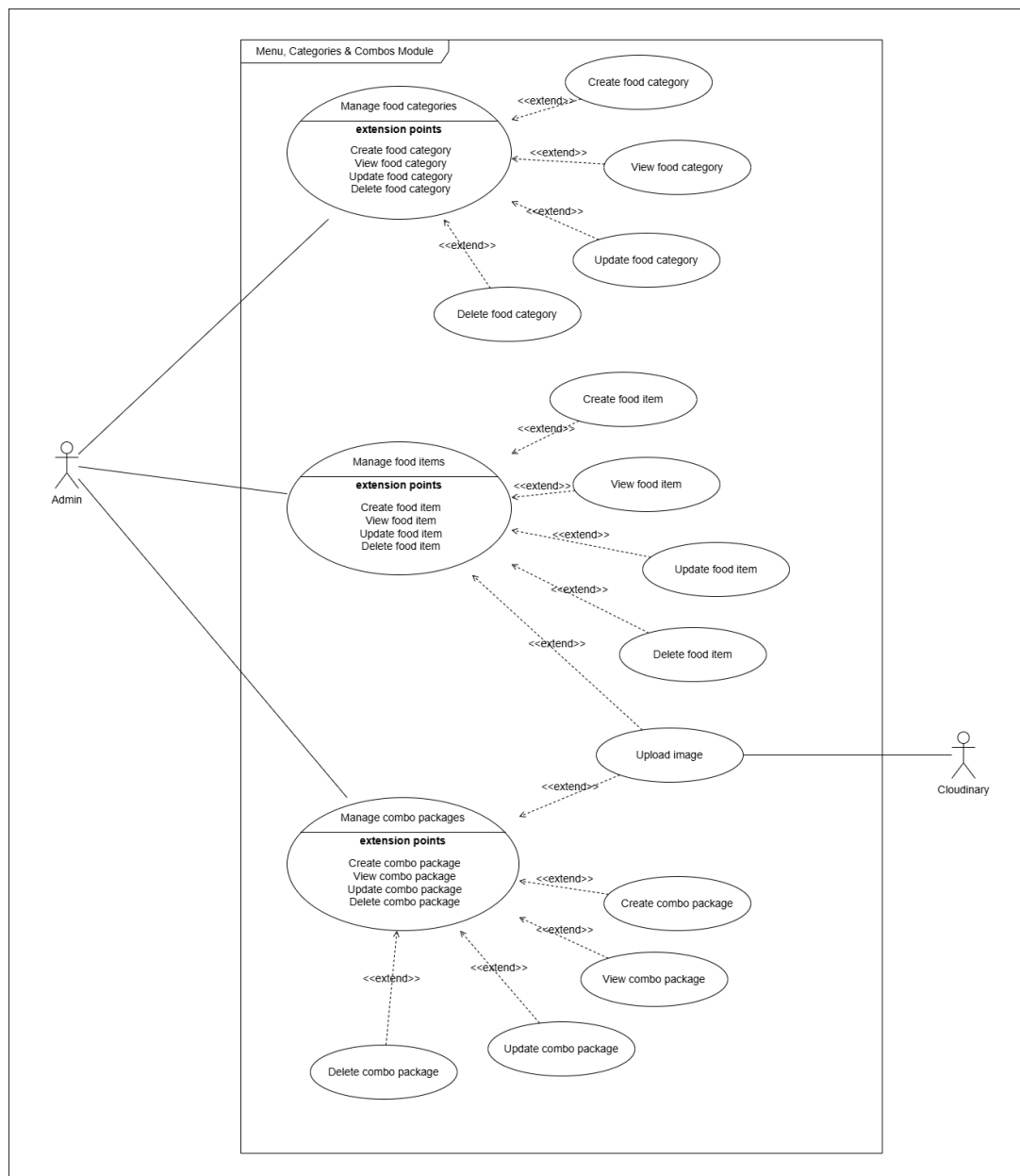


Figure I-10: Detailed Use Case Diagram for Managing Menu, Categories and Combos

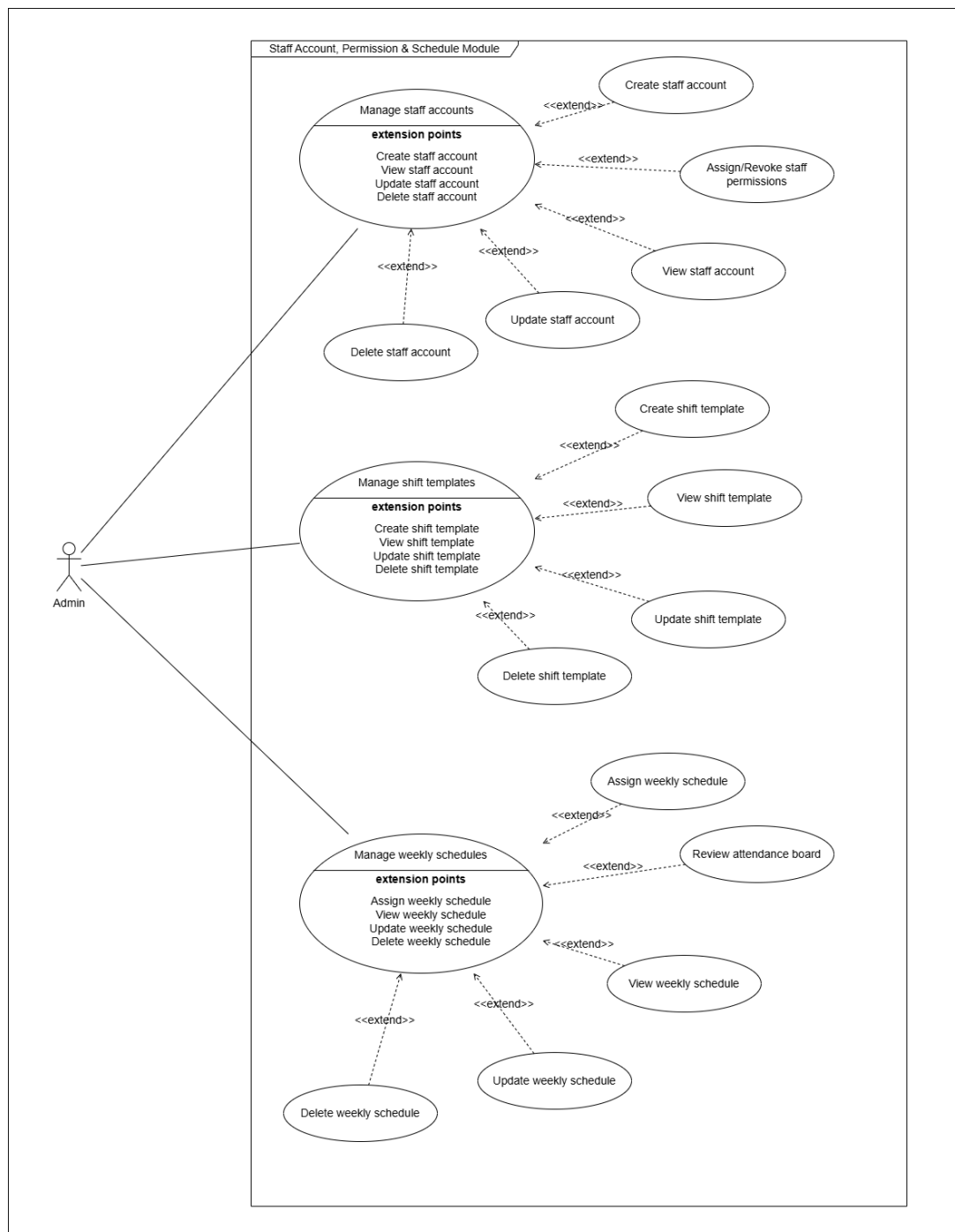


Figure I-11: Detailed Use Case Diagram for Staff Account, Permission and Schedule Management

I.5.2 Use Case Descriptions

This section specifies the detailed use cases of the CrownDine Intelligent Restaurant Management System. The use cases are grouped by customer, staff, kitchen, cashier, and admin workflows.

UC-01: Browse food menu by categories

ID and Name	UC-01 Browse food menu by categories
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customers to view the electronic menu grouped by categories such as Starters, Main Course, and Drinks.
Trigger	Customer selects the Menu section on the website.
Preconditions	None.
Postconditions	POST-1. Menu items are displayed correctly according to the chosen category.
Normal Flow	1. Customer navigates to the Menu page. 2. System retrieves active categories and corresponding food items from the database. 3. System displays category list and items under the default category. 4. Customer clicks different categories to filter the food list. 5. System refreshes menu items according to the selected category.
Alternative Flows	None.
Exceptions	None.
Priority	High
Business Rules	Only active categories and items are displayed to customers.

UC-02: View food item and combo details

ID and Name	UC-02 View food item and combo details
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customers to view detailed information of a dish or combo, including description, ingredients, price, image, and reviews.
Trigger	Customer clicks on a food item or combo card.
Preconditions	None.
Postconditions	POST-1. Detailed item/combo page is displayed.
Normal Flow	1. Customer selects a specific item or combo from the menu. 2. System queries detailed attributes of the selected item or combo sub-items. 3. System displays photos, details, price, and customer reviews.
Alternative Flows	None.
Exceptions	Item not found: system shows an error message and redirects back to Menu page.
Priority	High
Business Rules	Pricing shown must match the active base price configured by Admin.

UC-03: Search menu items by name

ID and Name	UC-03 Search menu items by name
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customers to search for food items or combos by entering keywords or names.
Trigger	Customer types in the search bar on the Menu page.
Preconditions	None.
Postconditions	POST-1. Menu list is filtered by the search query.
Normal Flow	1. Customer enters keywords into the search bar. 2. System queries database for items matching the search query using case-insensitive substring match. 3. System displays matching items or combos to the user.
Alternative Flows	None.
Exceptions	No matches found: system displays 'No items found matching your search'.
Priority	Medium
Business Rules	Search query must be sanitized to prevent SQL injection. Inactive menu items are excluded from search results.

UC-04: Manage favorite items

ID and Name	UC-04 Manage favorite items
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows logged-in customers to save food items to or remove them from their favorites list.
Trigger	Customer clicks the favorite heart icon on a food item card.
Preconditions	PRE-1. Customer is logged into the system.
Postconditions	POST-1. The favorite status of the item is updated in the database.
Normal Flow	1. Customer clicks the heart icon of an item. 2. System checks current status: if not in favorites, it adds item ID; if already in favorites, it removes it. 3. System updates the heart icon styling in UI.
Alternative Flows	None.
Exceptions	Customer is not logged in: system prompts customer to log in.
Priority	Low
Business Rules	Favorites are linked to customer account and persistent across sessions. Maximum of 50 favorite items per user.

UC-05: Submit feedback and rating on items

ID and Name	UC-05 Submit feedback and rating on items
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customers to submit reviews and star ratings on specific menu items they have ordered.
Trigger	Customer clicks Write Feedback on an item in their history.
Preconditions	PRE-1. Customer is logged in. PRE-2. Customer has ordered the item in the past.
Postconditions	POST-1. Feedback is saved and average rating is updated.
Normal Flow	1. Customer enters rating from 1 to 5 stars and review text, then clicks Submit. 2. System validates content and saves feedback in the database. 3. System recalculates average rating score of the item.
Alternative Flows	Customer can edit or update their existing feedback for the item.
Exceptions	None.
Priority	Medium
Business Rules	Customer must have ordered the item to be allowed to submit a review.

UC-06: View item feedbacks and reviews

ID and Name	UC-06 View item feedbacks and reviews
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customers to view reviews and ratings left by other diners on specific menu items.
Trigger	Customer opens the reviews section on a food detail page.
Preconditions	None.
Postconditions	POST-1. List of feedbacks, ratings, and comments are displayed.
Normal Flow	1. Customer views detail of an item. 2. System retrieves reviews, ratings, and comments associated with the item ID. 3. System displays review statistics and detailed review list.
Alternative Flows	None.
Exceptions	None.
Priority	Medium
Business Rules	Only approved or active reviews are displayed publicly. Reviews are sorted chronologically by default.

UC-07: Select reservation date, time, and guests

ID and Name	UC-07 Select reservation date, time, and guests
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Customer specifies date, time, and headcount to search for available seating slots.
Trigger	Customer clicks Book Table button.
Preconditions	PRE-1. Customer is logged in.
Postconditions	POST-1. Seating slot checking criteria is set.
Normal Flow	1. Customer opens Reservation Wizard. 2. Customer selects Date, Time Slot, and number of guests. 3. System checks opening hours from 09:00 to 22:00 and ensures date is not in the past. 4. System queries tables database to check availability.
Alternative Flows	None.
Exceptions	Selected date/time is in the past or outside business hours: system shows validation error.
Priority	High
Business Rules	Booking must be made within business hours from 09:00 to 22:00. Bookings cannot be made in the past.

UC-08: View table map and select preferred tables

ID and Name	UC-08 View table map and select preferred tables
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customer to view a 2D floor plan layout and select preferred tables to book.
Trigger	Customer advances to step 2 of booking.
Preconditions	PRE-1. Customer selected valid booking date, time, and guest count.
Postconditions	POST-1. Tables are selected and added to booking selection.
Normal Flow	1. System loads restaurant floor layouts. 2. System queries reservations matching selected date and time slot. 3. System renders table shapes, labels, and availability colors. 4. Customer clicks a green table matching their preference. 5. System marks selected table and highlights it.
Alternative Flows	None.
Exceptions	Table capacity is too small: system suggests a larger table.
Priority	High
Business Rules	Real-time table status synchronization is used. Guests must not exceed table capacity plus allowed tolerance.

UC-09: Manage reservation pre-order cart

ID and Name	UC-09 Manage reservation pre-order cart
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customer to add, edit quantity, or remove pre-ordered dishes in reservation cart.
Trigger	Customer interacts with pre-order cart interface.
Preconditions	PRE-1. Customer has selected booking tables.
Postconditions	POST-1. Pre-order items in reservation cart are updated.
Normal Flow	1. Customer browses items and adds them to booking cart. 2. System verifies item availability and updates quantities in session cart. 3. System recalculates total cart price.
Alternative Flows	None.
Exceptions	Item out of stock: system alerts customer that item cannot be ordered.
Priority	Medium
Business Rules	Quantity must be positive. Pre-order items can only be edited or removed before booking deposit is paid.

UC-10: Apply discount voucher to reservation

ID and Name	UC-10 Apply discount voucher to reservation
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customer to input a voucher code at reservation checkout to get a discount.
Trigger	Customer inputs code and clicks Apply.
Preconditions	PRE-1. Customer has active reservation order at step 4.
Postconditions	POST-1. Discount is applied to booking subtotal and final pay price is updated.
Normal Flow	1. Customer inputs voucher code. 2. System validates code by expiry date, active status, user constraints, and minimum order limit. 3. System calculates percentage or fixed discount and subtracts it from total bill.
Alternative Flows	None.
Exceptions	Voucher invalid, expired, or used: system displays error message.
Priority	High
Business Rules	Only one voucher can be applied per reservation order. Total bill must meet minimum spend requirements.

UC-11: Pay reservation deposit via PayOS

ID and Name	UC-11 Pay reservation deposit via PayOS
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	PayOS Payment Gateway
Description	Customer pays a booking deposit through PayOS gateway integration.
Trigger	Customer clicks Pay Now button.
Preconditions	PRE-1. Reservation details and table selection are completed.
Postconditions	POST-1. Reservation is registered with PENDING_PAYMENT status and PayOS redirect URL is generated.
Normal Flow	1. Customer clicks payment button. 2. System creates reservation with PENDING_PAYMENT status. 3. System communicates with PayOS API to generate checkout URL and QR code. 4. System redirects customer to PayOS checkout page. 5. Customer completes payment. 6. PayOS sends webhook and backend marks reservation CONFIRMED. 7. System redirects customer back to CrownDine success page.
Alternative Flows	Customer cancels payment: PayOS redirects to failure URL and system updates reservation status to FAILED.
Exceptions	PayOS API timeout: system displays payment service unavailable message.
Priority	High
Business Rules	Reservation hold lock on selected tables is released if payment is not completed within 15 minutes.

UC-12: Cancel pending/upcoming reservation

ID and Name	UC-12 Cancel pending/upcoming reservation
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customers to cancel their upcoming reservation from profile history.
Trigger	Customer clicks Cancel booking on an active booking card.
Preconditions	PRE-1. Booking has CONFIRMED status.
Postconditions	POST-1. Booking status is updated to CANCELLED and tables are freed.
Normal Flow	1. Customer selects booking cancellation. 2. System checks cancellation policy. 3. System updates reservation status to CANCELLED. 4. System sends email alert notifying successful cancellation.
Alternative Flows	None.
Exceptions	Late cancellation less than 2 hours before booking: system denies cancellation and directs user to support.
Priority	High
Business Rules	Cancellations are only allowed up to 2 hours before scheduled time slot.

UC-13: Register customer account

ID and Name	UC-13 Register customer account
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows new customers to create a CrownDine account.
Trigger	Customer submits registration form.
Preconditions	None.
Postconditions	POST-1. Temporary inactive account is created and OTP email is sent.
Normal Flow	1. Customer enters username, email, phone, password, and name. 2. System checks if username or email already exists. 3. System creates inactive user, hashes password, generates OTP, and sends email.
Alternative Flows	None.
Exceptions	Username or email already exists: system displays registration error.
Priority	High
Business Rules	Passwords must meet minimum security requirements.

UC-14: Verify registration via email OTP

ID and Name	UC-14 Verify registration via email OTP
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customers to verify email address using 6-digit OTP.
Trigger	Customer inputs OTP and clicks Verify.
Preconditions	PRE-1. Account registration initiated.
Postconditions	POST-1. Account status is activated.
Normal Flow	1. Customer checks email and enters OTP. 2. System validates OTP and expiry. 3. System sets customer user status to ACTIVE.
Alternative Flows	None.
Exceptions	Incorrect or expired OTP: system displays validation failure message.
Priority	High
Business Rules	OTP code is valid for exactly 5 minutes before expiring.

UC-15: Log in to the system

ID and Name	UC-15 Log in to the system
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer, Staff, Admin
Secondary Actors	Google Auth Service
Description	Allows registered users to log in to access their dashboards.
Trigger	User submits login credentials or clicks Google Login.
Preconditions	PRE-1. Account must be active.
Postconditions	POST-1. JWT access token is returned and user is redirected based on role.
Normal Flow	1. User enters credentials or uses Google Sign-In. 2. System validates user record in database. 3. System generates JWT tokens and maps authorities. 4. System redirects Admin to admin dashboard, Staff to staff dashboard, and Customer to Home.
Alternative Flows	None.
Exceptions	Invalid credentials or blocked account: login request is denied.
Priority	High
Business Rules	Access tokens expire in 1 hour; refresh tokens expire in 7 days. Failed logins lock account temporarily after 5 attempts.

UC-16: Log out of the system

ID and Name	UC-16 Log out of the system
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer, Staff, Admin
Secondary Actors	None
Description	Logs the user out and clears session tokens.
Trigger	User clicks Log Out button.
Preconditions	PRE-1. User is logged in.
Postconditions	POST-1. Refresh token is deleted and client storage is cleared.
Normal Flow	1. User clicks Log Out. 2. System deletes JWT refresh token record. 3. Client clears storage and redirects to Login or Home page.
Alternative Flows	None.
Exceptions	None.
Priority	High
Business Rules	Logged-out tokens are blacklisted to prevent reuse before expiration.

UC-17: View and update customer profile details

ID and Name	UC-17 View and update customer profile details
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customers to view and update personal name, phone number, and address.
Trigger	Customer edits profile form and clicks Save.
Preconditions	PRE-1. Customer is logged in.
Postconditions	POST-1. User details are updated in database.
Normal Flow	1. Customer opens profile. 2. Customer edits fields such as name and phone. 3. Customer clicks Save Changes. 4. System validates inputs and updates database record.
Alternative Flows	None.
Exceptions	Invalid phone format: system shows validation warning.
Priority	Medium
Business Rules	Phone number must consist of valid digits. Users can edit profile only while logged in.

UC-18: Update email address with OTP confirmation

ID and Name	UC-18 Update email address with OTP confirmation
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows customers to update email address by verifying it with OTP.
Trigger	Customer initiates email change in profile settings.
Preconditions	PRE-1. Customer is logged in.
Postconditions	POST-1. User email is changed after OTP verification.
Normal Flow	1. Customer inputs new email and clicks Send OTP. 2. System sends verification email with OTP. 3. Customer inputs received OTP and clicks Verify. 4. System validates OTP and updates email in database.
Alternative Flows	None.
Exceptions	Incorrect OTP: system rejects update and prompts user to check code.
Priority	Medium
Business Rules	New email must not already exist. OTP is single-use and invalid after verification attempt.

UC-19: Upload profile avatar

ID and Name	UC-19 Upload profile avatar
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	Cloudinary Service
Description	Allows customers to upload an image file as profile picture.
Trigger	Customer selects avatar photo and clicks Upload.
Preconditions	PRE-1. Customer is logged in.
Postconditions	POST-1. Avatar is uploaded and URL is updated in user record.
Normal Flow	1. Customer clicks Change Avatar and selects image file. 2. System uploads file to Cloudinary. 3. Cloudinary returns image URL. 4. System updates avatar URL in user record. 5. System displays new avatar in UI.
Alternative Flows	None.
Exceptions	File too large or invalid format: system shows warning.
Priority	Low
Business Rules	Supported formats: JPG, PNG, GIF. Max file size: 2MB.

UC-20: Change account password

ID and Name	UC-20 Change account password
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer, Staff, Admin
Secondary Actors	None
Description	Allows logged-in users to update account passwords.
Trigger	User submits Change Password form.
Preconditions	PRE-1. User is logged in.
Postconditions	POST-1. Password is updated in database.
Normal Flow	1. User enters current password, new password, and confirmation. 2. System validates current password. 3. System hashes new password and updates database record.
Alternative Flows	None.
Exceptions	Incorrect current password: password update is rejected.
Priority	Medium
Business Rules	New password must differ from current password and must be hashed using BCrypt.

UC-21: View booking and order history

ID and Name	UC-21 View booking and order history
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows Customer to perform the CrownDine workflow: View booking and order history.
Trigger	Customer selects or submits the View booking and order history action.
Preconditions	PRE-1. Customer is logged in and authorized.
Postconditions	POST-1. View booking and order history workflow is completed and system state is updated.
Normal Flow	1. Customer opens the View booking and order history function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-22: Exchange loyalty points for vouchers

ID and Name	UC-22 Exchange loyalty points for vouchers
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows Customer to perform the CrownDine workflow: Exchange loyalty points for vouchers.
Trigger	Customer selects or submits the Exchange loyalty points for vouchers action.
Preconditions	PRE-1. Customer is logged in and authorized.
Postconditions	POST-1. Exchange loyalty points for vouchers workflow is completed and system state is updated.
Normal Flow	1. Customer opens the Exchange loyalty points for vouchers function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-23: Chat with AI booking chatbot

ID and Name	UC-23 Chat with AI booking chatbot
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	Gemini AI API
Description	Allows Customer to perform the CrownDine workflow: Chat with AI booking chatbot.
Trigger	Customer selects or submits the Chat with AI booking chatbot action.
Preconditions	PRE-1. Customer is logged in and authorized.
Postconditions	POST-1. Chat with AI booking chatbot workflow is completed and system state is updated.
Normal Flow	1. Customer opens the Chat with AI booking chatbot function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-24: Navigate booking steps via AI links

ID and Name	UC-24 Navigate booking steps via AI links
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Customer
Secondary Actors	None
Description	Allows Customer to perform the CrownDine workflow: Navigate booking steps via AI links.
Trigger	Customer selects or submits the Navigate booking steps via AI links action.
Preconditions	PRE-1. Customer is logged in and authorized.
Postconditions	POST-1. Navigate booking steps via AI links workflow is completed and system state is updated.
Normal Flow	1. Customer opens the Navigate booking steps via AI links function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-25: View reservation list

ID and Name	UC-25 View reservation list
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Staff
Secondary Actors	None
Description	Allows Staff to perform the CrownDine workflow: View reservation list.
Trigger	Staff selects or submits the View reservation list action.
Preconditions	PRE-1. Staff is logged in and authorized.
Postconditions	POST-1. View reservation list workflow is completed and system state is updated.
Normal Flow	1. Staff opens the View reservation list function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-26: Confirm customer check-in

ID and Name	UC-26 Confirm customer check-in
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Staff
Secondary Actors	None
Description	Allows Staff to perform the CrownDine workflow: Confirm customer check-in.
Trigger	Staff selects or submits the Confirm customer check-in action.
Preconditions	PRE-1. Staff is logged in and authorized.
Postconditions	POST-1. Confirm customer check-in workflow is completed and system state is updated.
Normal Flow	1. Staff opens the Confirm customer check-in function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-27: Update reservation table assignment

ID and Name	UC-27 Update reservation table assignment
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Staff
Secondary Actors	None
Description	Allows Staff to perform the CrownDine workflow: Update reservation table assignment.
Trigger	Staff selects or submits the Update reservation table assignment action.
Preconditions	PRE-1. Staff is logged in and authorized.
Postconditions	POST-1. Update reservation table assignment workflow is completed and system state is updated.
Normal Flow	1. Staff opens the Update reservation table assignment function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-28: Cancel reservation by staff

ID and Name	UC-28 Cancel reservation by staff
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Staff
Secondary Actors	None
Description	Allows Staff to perform the CrownDine workflow: Cancel reservation by staff.
Trigger	Staff selects or submits the Cancel reservation by staff action.
Preconditions	PRE-1. Staff is logged in and authorized.
Postconditions	POST-1. Cancel reservation by staff workflow is completed and system state is updated.
Normal Flow	1. Staff opens the Cancel reservation by staff function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-29: Create walk-in dine-in order

ID and Name	UC-29 Create walk-in dine-in order
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Staff (Cashier)
Secondary Actors	None
Description	Allows Staff (Cashier) to perform the CrownDine workflow: Create walk-in dine-in order.
Trigger	Staff (Cashier) selects or submits the Create walk-in dine-in order action.
Preconditions	PRE-1. Staff (Cashier) is logged in and authorized.
Postconditions	POST-1. Create walk-in dine-in order workflow is completed and system state is updated.
Normal Flow	1. Staff (Cashier) opens the Create walk-in dine-in order function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-30: Map customer to order by phone number

ID and Name	UC-30 Map customer to order by phone number
Created By	Lê Trần Bảo Phúc
Date Created	09/06/2026
Primary Actor	Staff (Cashier)
Secondary Actors	None
Description	Allows Staff (Cashier) to perform the CrownDine workflow: Map customer to order by phone number.
Trigger	Staff (Cashier) selects or submits the Map customer to order by phone number action.
Preconditions	PRE-1. Staff (Cashier) is logged in and authorized.
Postconditions	POST-1. Map customer to order by phone number workflow is completed and system state is updated.
Normal Flow	1. Staff (Cashier) opens the Map customer to order by phone number function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-31: Append items/combo to table order

ID and Name	UC-31 Append items/combos to table order
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Staff (Cashier)
Secondary Actors	None
Description	Allows Staff (Cashier) to perform the CrownDine workflow: Append items/combos to table order.
Trigger	Staff (Cashier) selects or submits the Append items/combos to table order action.
Preconditions	PRE-1. Staff (Cashier) is logged in and authorized.
Postconditions	POST-1. Append items/combos to table order workflow is completed and system state is updated.
Normal Flow	1. Staff (Cashier) opens the Append items/combos to table order function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-32: Apply discount voucher to table bill

ID and Name	UC-32 Apply discount voucher to table bill
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Staff (Cashier)
Secondary Actors	None
Description	Allows Staff (Cashier) to perform the CrownDine workflow: Apply discount voucher to table bill.
Trigger	Staff (Cashier) selects or submits the Apply discount voucher to table bill action.
Preconditions	PRE-1. Staff (Cashier) is logged in and authorized.
Postconditions	POST-1. Apply discount voucher to table bill workflow is completed and system state is updated.
Normal Flow	1. Staff (Cashier) opens the Apply discount voucher to table bill function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-33: Process table payment via PayOS or Cash

ID and Name	UC-33 Process table payment via PayOS or Cash
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Staff (Cashier)
Secondary Actors	PayOS Gateway
Description	Allows Staff (Cashier) to perform the CrownDine workflow: Process table payment via PayOS or Cash.
Trigger	Staff (Cashier) selects or submits the Process table payment via PayOS or Cash action.
Preconditions	PRE-1. Staff (Cashier) is logged in and authorized.
Postconditions	POST-1. Process table payment via PayOS or Cash workflow is completed and system state is updated.
Normal Flow	1. Staff (Cashier) opens the Process table payment via PayOS or Cash function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-34: Split bill / Merge tables

ID and Name	UC-34 Split bill / Merge tables
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Staff (Cashier)
Secondary Actors	None
Description	Allows Staff (Cashier) to perform the CrownDine workflow: Split bill / Merge tables.
Trigger	Staff (Cashier) selects or submits the Split bill / Merge tables action.
Preconditions	PRE-1. Staff (Cashier) is logged in and authorized.
Postconditions	POST-1. Split bill / Merge tables workflow is completed and system state is updated.
Normal Flow	1. Staff (Cashier) opens the Split bill / Merge tables function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	Medium
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-35: Print invoice receipt

ID and Name	UC-35 Print invoice receipt
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Staff (Cashier)
Secondary Actors	None
Description	Allows Staff (Cashier) to perform the CrownDine workflow: Print invoice receipt.
Trigger	Staff (Cashier) selects or submits the Print invoice receipt action.
Preconditions	PRE-1. Staff (Cashier) is logged in and authorized.
Postconditions	POST-1. Print invoice receipt workflow is completed and system state is updated.
Normal Flow	1. Staff (Cashier) opens the Print invoice receipt function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-36: View pending kitchen order queue

ID and Name	UC-36 View pending kitchen order queue
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Kitchen Staff
Secondary Actors	None
Description	Allows Kitchen Staff to perform the CrownDine workflow: View pending kitchen order queue.
Trigger	Kitchen Staff selects or submits the View pending kitchen order queue action.
Preconditions	PRE-1. Kitchen Staff is logged in and authorized.
Postconditions	POST-1. View pending kitchen order queue workflow is completed and system state is updated.
Normal Flow	1. Kitchen Staff opens the View pending kitchen order queue function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-37: Group items into batches and Update preparation status

ID and Name	UC-37 Group items into batches and Update preparation status
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Kitchen Staff
Secondary Actors	None
Description	Allows Kitchen Staff to perform the CrownDine workflow: Group items into batches and Update preparation status.
Trigger	Kitchen Staff selects or submits the Group items into batches and Update preparation status action.
Preconditions	PRE-1. Kitchen Staff is logged in and authorized.
Postconditions	POST-1. Group items into batches and Update preparation status workflow is completed and system state is updated.
Normal Flow	1. Kitchen Staff opens the Group items into batches and Update preparation status function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-38: View assigned weekly shifts schedule

ID and Name	UC-38 View assigned weekly shifts schedule
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Staff
Secondary Actors	None
Description	Allows Staff to perform the CrownDine workflow: View assigned weekly shifts schedule.
Trigger	Staff selects or submits the View assigned weekly shifts schedule action.
Preconditions	PRE-1. Staff is logged in and authorized.
Postconditions	POST-1. View assigned weekly shifts schedule workflow is completed and system state is updated.
Normal Flow	1. Staff opens the View assigned weekly shifts schedule function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-39: Perform shifts attendance clock-in/out

ID and Name	UC-39 Perform shifts attendance clock-in/out
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Staff
Secondary Actors	None
Description	Allows Staff to perform the CrownDine workflow: Perform shifts attendance clock-in/out.
Trigger	Staff selects or submits the Perform shifts attendance clock-in/out action.
Preconditions	PRE-1. Staff is logged in and authorized.
Postconditions	POST-1. Perform shifts attendance clock-in/out workflow is completed and system state is updated.
Normal Flow	1. Staff opens the Perform shifts attendance clock-in/out function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-40: Manage floors and areas

ID and Name	UC-40 Manage floors and areas
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Manage floors and areas.
Trigger	Admin selects or submits the Manage floors and areas action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Manage floors and areas workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Manage floors and areas function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	Medium
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-41: Design floor layout and table configuration

ID and Name	UC-41 Design floor layout and table configuration
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Design floor layout and table configuration.
Trigger	Admin selects or submits the Design floor layout and table configuration action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Design floor layout and table configuration workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Design floor layout and table configuration function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-42: Manage food categories

ID and Name	UC-42 Manage food categories
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Manage food categories.
Trigger	Admin selects or submits the Manage food categories action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Manage food categories workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Manage food categories function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-43: Manage food items

ID and Name	UC-43 Manage food items
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	Cloudinary
Description	Allows Admin to perform the CrownDine workflow: Manage food items.
Trigger	Admin selects or submits the Manage food items action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Manage food items workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Manage food items function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-44: Manage combo packages

ID and Name	UC-44 Manage combo packages
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	Cloudinary
Description	Allows Admin to perform the CrownDine workflow: Manage combo packages.
Trigger	Admin selects or submits the Manage combo packages action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Manage combo packages workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Manage combo packages function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-45: Create discount vouchers

ID and Name	UC-45 Create discount vouchers
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Create discount vouchers.
Trigger	Admin selects or submits the Create discount vouchers action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Create discount vouchers workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Create discount vouchers function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-46: Edit/Delete discount vouchers

ID and Name	UC-46 Edit/Delete discount vouchers
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Edit/Delete discount vouchers.
Trigger	Admin selects or submits the Edit/Delete discount vouchers action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Edit/Delete discount vouchers workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Edit/Delete discount vouchers function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	Medium
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-47: Assign vouchers to users

ID and Name	UC-47 Assign vouchers to users
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Assign vouchers to users.
Trigger	Admin selects or submits the Assign vouchers to users action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Assign vouchers to users workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Assign vouchers to users function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	Medium
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-48: Create and manage staff accounts

ID and Name	UC-48 Create and manage staff accounts
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Create and manage staff accounts.
Trigger	Admin selects or submits the Create and manage staff accounts action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Create and manage staff accounts workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Create and manage staff accounts function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-49: Assign/revoke staff permissions and status

ID and Name	UC-49 Assign/revoke staff permissions and status
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Assign/revoke staff permissions and status.
Trigger	Admin selects or submits the Assign/revoke staff permissions and status action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Assign/revoke staff permissions and status workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Assign/revoke staff permissions and status function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-50: Configure shifts template

ID and Name	UC-50 Configure shifts template
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Configure shifts template.
Trigger	Admin selects or submits the Configure shifts template action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Configure shifts template workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Configure shifts template function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	Medium
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-51: Assign weekly work schedule to staff

ID and Name	UC-51 Assign weekly work schedule to staff
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Assign weekly work schedule to staff.
Trigger	Admin selects or submits the Assign weekly work schedule to staff action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Assign weekly work schedule to staff workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Assign weekly work schedule to staff function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-52: Review weekly attendance board

ID and Name	UC-52 Review weekly attendance board
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Review weekly attendance board.
Trigger	Admin selects or submits the Review weekly attendance board action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Review weekly attendance board workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Review weekly attendance board function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	Medium
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-53: View interactive dashboard sales graphs

ID and Name	UC-53 View interactive dashboard sales graphs
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: View interactive dashboard sales graphs.
Trigger	Admin selects or submits the View interactive dashboard sales graphs action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. View interactive dashboard sales graphs workflow is completed and system state is updated.
Normal Flow	1. Admin opens the View interactive dashboard sales graphs function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-54: Export reports to Excel

ID and Name	UC-54 Export reports to Excel
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	None
Description	Allows Admin to perform the CrownDine workflow: Export reports to Excel.
Trigger	Admin selects or submits the Export reports to Excel action.
Preconditions	PRE-1. Admin is logged in and authorized.
Postconditions	POST-1. Export reports to Excel workflow is completed and system state is updated.
Normal Flow	1. Admin opens the Export reports to Excel function. 2. System validates role, input data, and related business constraints. 3. System performs the requested workflow and updates database records when required. 4. System refreshes the interface and shows the result to the actor.
Alternative Flows	None.
Exceptions	Invalid data, unavailable records, or access denial: system displays an error message.
Priority	High
Business Rules	Only authorized actors can perform this use case. All changes are validated and persisted consistently.

UC-55: Chat with Gemini AI for sales metrics and business health reports

ID and Name	UC-55 Chat with Gemini AI for sales metrics and business health reports
Created By	Lê Trần Bảo Phúc
Date Created	10/06/2026
Primary Actor	Admin
Secondary Actors	Gemini AI API
Description	Allows Admin to interact with Gemini AI chatbot to query real-time revenue stats, top-selling combos, and business suggestions.
Trigger	Admin submits analytical queries to AI Chatbot.
Preconditions	PRE-1. Admin is logged in.
Postconditions	POST-1. Streaming text response from Gemini AI is displayed.
Normal Flow	1. Admin inputs analytical query about revenue and top-selling combos. 2. System routes query to AI Admin Chat Service. 3. AI detects and triggers secured tools such as getRevenueReport and getTopSellingCombos. 4. AI compiles analytical reply and streams response to admin chatbot window.
Alternative Flows	None.
Exceptions	Rate limits or API failure: system shows alert.
Priority	High
Business Rules	Tool calling functions are secured and only accessible if user role is ADMIN.

I.5.3 Activity Diagrams

This section should be completed by the project team with project-specific details, diagrams, tables, and references for the Crowndine Restaurant Management System.

I.6 Data Requirements

Core data entities include User, Customer, Staff, Menu Item, Category, Reservation, Table, Order, Order Detail, Payment, Feedback, and Notification.

II Analysis Models

II.1 Interaction Diagrams

This section should be completed by the project team with project-specific details, diagrams, tables, and references for the Crowndine Restaurant Management System.

II.2 State Diagram

The main order lifecycle may include Created, Confirmed, Preparing, Ready, Served, Completed, and Cancelled states. State transitions must be triggered only by authorized users or valid system events.

III Design Specification

III.1 Integrated Communication Diagrams

This section should be completed by the project team with project-specific details, diagrams, tables, and references for the Crowndine Restaurant Management System.

III.2 System High-Level Design

The system follows a layered architecture consisting of presentation, application service, domain logic, data access, and persistence layers. Each layer separates responsibilities to improve maintainability and testability.

III.3 Component & Package Diagram

This section should be completed by the project team with project-specific details, diagrams, tables, and references for the Crowndine Restaurant Management System.

III.4 Detailed Design

After the high-level architectural design, the classes identified in the analysis model are transformed into design classes. The operations of each class are derived from both the static model and the dynamic model. Following the information-hiding principle, the design classes are categorized into three stereotypes: boundary, control, and entity classes. In the design model, each entity is represented by two classes: a data abstraction class that encapsulates the entity's attributes and relationships, and a database wrapper class, implemented as a Spring Data JPA Repository, that encapsulates the persistence operations and hides the underlying database access mechanism.

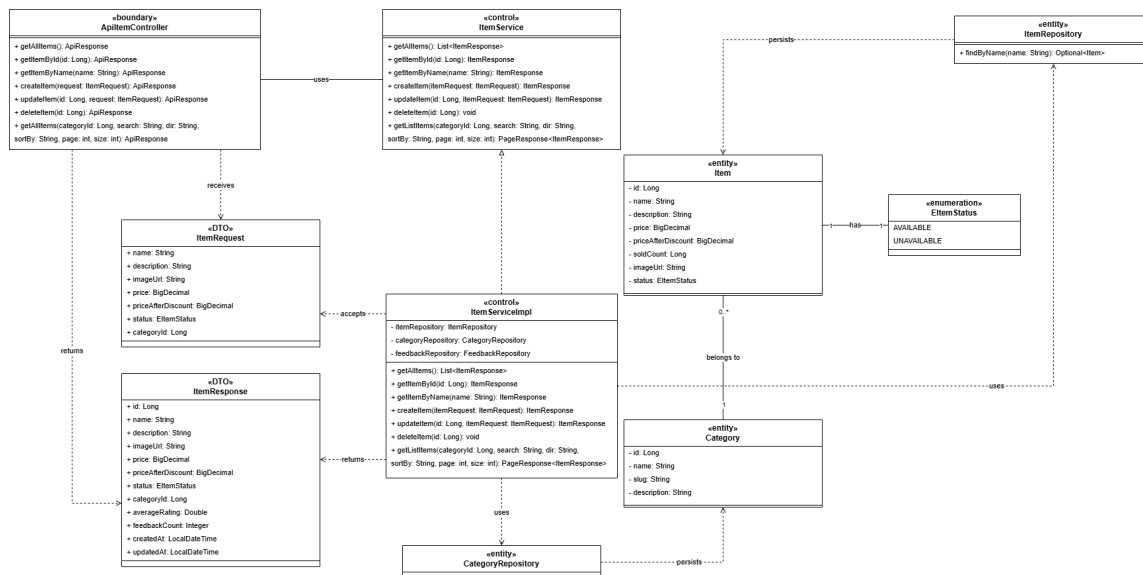


Figure III-8: Detailed Design Class Diagram for Manage Item Use Case

Figure III-8 illustrates the detailed design classes for the Manage Item use case, supporting the following operations: Get Item List, Get Item by Id, Get Item by Name, Create Item, Update Item and Delete Item. These classes are derived from the CrownDine high-level architecture presented in Section III.2. Their operations are derived from the sequence of messages exchanged among the boundary, control, and entity classes in the corresponding sequence diagrams.

ApiController is the boundary class responsible for receiving HTTP requests from clients and returning standardized **ApiResponse** objects. **ItemService** as an interface and **ItemServiceImpl** as its implementation are the control classes that coordinate the business logic for the Manage Item use case, including processing requests, resolving the associated category and delegating persistence operations.

The entity `Item` is represented by `ItemRepository`, which encapsulates the persistence operations, including the custom query method `findByName`, the standard persistence operations inherited from `JpaRepository<Item, Long>` and dynamic filtering via `JpaSpecificationExecutor<Item>`. The `Item` class encapsulates the item's attributes such as `name`, `price`, `priceAfterDiscount`, `soldCount`, `imageUrl` and `status`. Similarly, because each item belongs to exactly one category, the supporting entity `Category` is represented by `CategoryRepository` and `Category`.

The DTO classes `ItemRequest` and `ItemResponse` transfer data between the boundary and control layers while isolating the internal entity model from external clients. Finally, the enumer-

ation `EItemStatus`, including `AVAILABLE` and `UNAVAILABLE`, defines the possible status values of an item.

III.5 Database Design

The database design shall include an entity relationship diagram, table schemas, primary keys, foreign keys, constraints, indexes and sample records where applicable.

IV Implementation

IV.1 Project Architecture

The implementation should organize modules by responsibility, such as user management, menu management, reservation management, order management, payment integration, notification and reporting.

IV.2 Mapping Class Diagrams & Interaction Diagrams to Code

Class diagrams shall be mapped to source code classes, interfaces, models, repositories, controllers, and services. Interaction diagrams shall be mapped to method calls, API flows and transaction boundaries.

V Applying Alternative Architecture Patterns

Assuming that the functional requirements remain unchanged, the system is now being expanded to incorporate additional characteristics related to scalability and reusability. This enhancement allows the system to support the integration of new user-facing applications, including the potential development of a mobile app.

V.1 Applying the Service-Oriented Architecture

NF-05 Reusability: The system should be built so that the same code and features can be reused across different platforms, allowing the same backend functionality to serve both the Web and Mobile applications without duplicating business logic. This saves time and effort when adding new platforms or updating existing ones. To satisfy this requirement, the architecture is redesigned using a service-oriented approach.

The current CrownDine architecture is a single monolithic Spring Boot backend, which does not fully satisfy NF-05. Any new client platform, such as the upcoming mobile application, would have to depend on the same tightly-coupled backend with no clear service boundary. To address this, the backend is restructured into six coarse-grained service components: Auth & User Service, Menu Service, Reservation & Table Service, Order & Payment Service, Staff & Workforce Service and Notification & AI Service. Each service is deployed as an independent Spring Boot application artifact within its own execution environment.

The system exposes a single REST API contract through the API Gateway Server, implemented using Spring Cloud Gateway. Both the Web Browser, implemented as a React single-page application, and the Mobile App, implemented using React Native, consume this contract over HTTPS, while the gateway forwards each request over HTTP to the appropriate backend service. Because both platforms rely on the exact same contract and the same set of services, no backend logic needs to be duplicated or re-implemented when a new client platform is introduced. This directly fulfills NF-05 Reusability.

All six services continue to share a single MySQL database using the **crowndine** schema, accessed through JPA/Hibernate over JDBC. This design intentionally adopts a shared-database approach instead of the database-per-service pattern commonly adopted in microservice architectures in order to simplify data consistency and reduce migration complexity. Therefore, the proposed architecture reflects a service-oriented style rather than a fully decentralized microservices style.

In addition, three services integrate with external cloud providers. The Order & Payment Service connects to PayOS for payment processing, the Menu Service connects to Cloudinary for media storage, and the Notification & AI Service connects to both Gemini AI for the chatbot feature and an external Email/SMS provider for notification delivery. Each external service is reached over HTTPS.

This restructuring transforms the original monolithic architecture into a distributed service-oriented architecture. By exposing a common REST API contract through the API Gateway, the backend services can be reused across multiple client platforms without duplicating business logic, thereby satisfying NF-05 Reusability. In addition, the architecture provides a clearer separation of business capabilities, improved modularity, easier maintenance, and greater flexibility for future system evolution. The resulting software architecture is illustrated in Figure V-1.

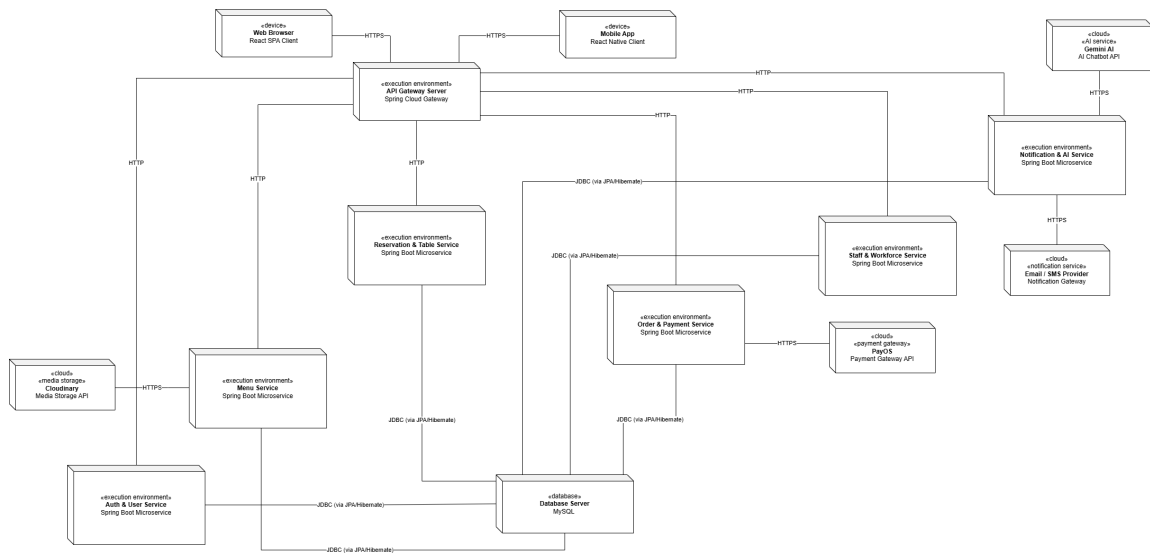


Figure V-1: Deployment Diagram for the CrownDine System Architecture Based on Service-Oriented Architecture

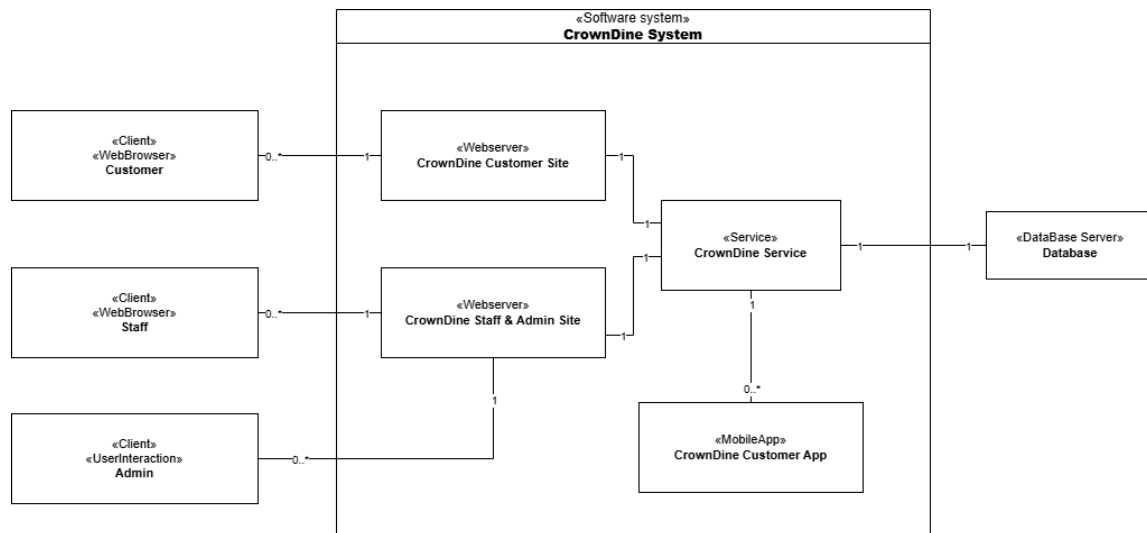


Figure V-2: Class Diagram for the System Architecture Based on Service-Oriented Architecture